Summary: A Comparative Evaluation of RAG Architectures for Cross-Domain LLM Applications: Design, Implementation, and Assessment

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Summary by Syed Ukkashah · 23K-0055 · Assignment Beta, Question 5

1. Problem and motivation

Large language models generalise well, but their knowledge is frozen at training time, and they sometimes invent specifics. In finance and e-commerce that is unacceptable: answers depend on exact figures, currencies and dates, policies change often, and regulators expect every claim to be traceable to a source. Retrieval-Augmented Generation (RAG) addresses this by retrieving evidence at query time and conditioning the answer on it. Four retrieval paradigms dominate practice (sparse, dense, hybrid and fusion), but earlier work evaluates them on different data, models and metrics. Practitioners therefore lack a like-for-like comparison of their accuracy, speed and auditability. The paper sets out to provide that comparison with a reproducible protocol.

Research questions and contributions. The study asks five research questions: (1) how the four pipelines compare on faithfulness and recall when the generator is held fixed; (2) how sensitive results are to retrieval depth, chunking and index tuning; (3) which paradigm gives the best utility under latency constraints; (4) how they differ in citation coverage and traceability ("auditability"); and (5) whether the advantages hold across two hosting set-ups and query difficulty levels. The claimed contributions are a pre-registered evaluation protocol with bootstrap confidence intervals and paired tests, a claim-level auditability metric, instrumented LangChain templates for each paradigm, and decision-support visuals for choosing a pipeline under budget and service-level constraints.

2. The four architectures compared

Pipeline	How it retrieves (paper's configuration)	Strength	Weakness
Sparse	BM25 keyword scoring (k1 = 1.2, b = 0.75); 512-token chunks; top-10, optional cross-encoder re-rank to 5	Every score is inspectable; millisecond lookups on a CPU; precise for standardised wording	Misses paraphrases and other languages; new tickers and SKUs need re-indexing
Dense	all-mpnet-base-v2 embeddings (768-d) in an HNSW index; 384-token chunks; top-20, MiniLM cross-encoder to 5	Finds evidence with no word overlap; works across languages	Similarity scores are opaque to auditors; embedding the corpus is costly and must be redone when it changes
Hybrid	BM25 and dense retrieval run in parallel (15 + 15 passages) and merged with reciprocal-rank fusion, then re-ranked to 8	Precision of BM25 plus recall of embeddings; one track compensates when the other fails	Two indexes to build, tune and keep in sync
Fusion	An LLM writes m = 4 sub-queries, each retrieving 10 passages (or Fusion-in-Decoder over up to 28); 256-token chunks; re-rank to 10	Gathers evidence from several angles; strongest factual coverage	Slowest; larger prompts; harder to trace which passage produced which claim

3. Experimental setup

Data. A finance corpus (3,214 SEC 10-K/10-Q filings, investor-relations pages and central-bank circulars, June 2019 – June 2025, 92.7 M tokens) and an e-commerce corpus (4,836 product, pricing, returns-policy and help-centre passages in English and Turkish, 68.3 M tokens) were cleaned, de-duplicated and normalised (currencies, dates). A separate evidence pool of 7,150 passages has source IDs and timestamps.

Queries. 500 finance and e-commerce questions, stratified by domain and difficulty. 25% are deliberately unanswerable (built with time shifts and entity perturbations), so a system cannot score well simply by refusing. Annotator agreement on a 20% sample was substantial (Cohen's $\kappa = 0.77$ for answerability, 0.69 for domain, 0.74 for difficulty).

Generators and metrics. Cloud GPT-4o and an on-premises Llama-3-8B-Instruct (4-bit GGUF, llama.cpp on a MacBook Pro M3), both with temperature 0.2 and no fine-tuning. Within each comparison the generator and prompts are fixed, so differences can be attributed to retrieval. Metrics: the RAGAS scores (faithfulness = share of answer claims entailed by a cited passage; answer relevance; context precision; context recall at k), plus auditability (share of claims with a valid, recent citation) and end-to-end latency. Results are reported with 95% bootstrap confidence intervals and paired Wilcoxon tests with Holm correction.

4. Key results

Pipeline	Faithfulness	Answer relevance	Context precision	Context recall	Auditability	Median latency
Zero-shot (no retrieval)	0.58	0.61	0.42	0.38	0.35	680 ms
Sparse (BM25)	0.70	0.76	0.66	0.62	0.71	880 ms
Dense (ANN)	0.74	0.80	0.63	0.78	0.73	1,020 ms
Hybrid (RRF + re-rank)	0.80	0.83	0.69	0.75	0.79	1,110 ms
Fusion (expansion / FiD)	0.82	0.84	0.71	0.77	0.81	1,230 ms

Values are means from the paper's Table 3 (95% CIs ±0.02–0.04). Higher is better, except latency.

- **Retrieval always helps.** Without retrieval, GPT-4o invented monetary figures or out-of-date regulatory thresholds in roughly one third of answers; every RAG pipeline cut this to single digits. Faithfulness rises from 0.58 to between 0.70 and 0.82, and auditability roughly doubles (0.35 to 0.71–0.81).
- **Fusion is the most accurate.** Fusion scores highest on faithfulness, relevance, precision and auditability, but is the slowest (1.23 s median, 550 ms more than no retrieval).
- **Hybrid is the best balance.** Hybrid comes within 0.02 of fusion on nearly every metric and is 120 ms faster; the authors call it the best overall trade-off.
- **Dense and sparse sit at the extremes.** Dense has the highest context recall (0.78), because it matches meaning rather than words. Sparse is the fastest RAG pipeline (880 ms), but its recall (0.62) suffers on paraphrased and multilingual questions.
- **Source quality matters.** Answers that cite primary sources (e.g. EDGAR filings) and recent passages are consistently more faithful, so source authority and freshness act as moderators of RAG accuracy.

5. Recommendations, limitations and future work

Deployment guidance. Choose sparse or hybrid when auditability and strict latency dominate; dense or hybrid for paraphrase-heavy or multilingual interactive traffic; and fusion, ideally with citation filtering, when maximum factual coverage matters and batch or offline latency is acceptable.

Limitations (stated by the authors). Only two domains are tested; results may change with other generators, encoders, indexes and prompt budgets; RAGAS-style metrics are automatic proxies for human judgement; and cost and carbon are not measured. Future work: dynamic indexing, retrieval-aware decoding and joint training, long-context generators with claim-level citations, and cross-domain replication with human judges.

6. Critical assessment

The paper is a useful practitioner's guide: the four designs are explained with concrete examples, the configurations are fully specified, and including unanswerable questions is a good safeguard. However, several weaknesses limit how far the conclusions can be trusted:

- **The text contradicts its own table.** The text and radar plot say sparse retrieval has the best auditability and context precision, yet Table 3 ranks sparse lowest of the four RAG pipelines on both (0.71 and 0.66).
- **The headline gap is within noise.** Fusion's faithfulness (0.82 ± 0.02) overlaps hybrid's (0.80 ± 0.02), and the paired-test p-values and effect sizes are deferred to a supplement, so "fusion is most faithful" is not clearly supported. Hybrid is arguably the real winner.
- **Several research questions go unanswered.** Results are given as one number per pipeline, with no split by GPT-4o vs Llama, domain or difficulty, even though RQ5 asks exactly that. The sensitivity ablations and Pareto frontiers promised for RQ2 are not shown.
- **Limited verifiability.** Answer keys are not released, scoring is fully automatic, and the code and data are available only on request, so the results cannot be checked independently. Editing residue such as "addressing the reviewers' requests" also remains in the published text.

Takeaway. Retrieval clearly and substantially reduces hallucination in fact-critical domains, and hybrid retrieval is the safest default. Finer distinctions between the designs need the stratified results the paper does not yet report.